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Recognition of Laryngeal DiseasesUsing Artificial Intelligence

(Graduation Project 1) prepared to obtain the degree of B.Sc. **in the Department of Artificial Intelligence Engineering and Data Science**

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled
"Recognition of Laryngeal Diseases Using Artificial Intelligence"
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was carried out under our supervision at the Faculty of Artificial Intelligence Engineering in partial fulfillment of the requirements for the degree of **B.Sc.** in the Department of Artificial Intelligence Engineering and Data Science.

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Acknowledgements

This project was carried out at the Faculty of Artificial Intelligence Engineering, Syrian Private University. We would like to express our sincere gratitude to our supervisors, Dr. Mouhib Al-noukari, Dr. Kadan Al-joumaa and Eng. Anas Abd Abdulaziz, for their continuous guidance, insightful feedback, and unwavering support throughout the course of this research.

We extend our appreciation to the Department of Artificial Intelligence Engineering and Data Science for providing the academic environment and resources essential for this work. We also thank all colleagues and faculty members who offered assistance and constructive discussions during the development of this project.

Dedication

To our families, whose constant encouragement and patience made this journey possible. To every physician and patient who awaits more accurate and accessible diagnostic tools — this work is for you.

Abstract

Laryngeal diseases, including carcinoma, leukoplakia, papilloma, and Reinke's edema, represent a clinically significant disease spectrum whose accurate and timely diagnosis is critical for patient outcomes. Manual interpretation of laryngoscopic images is time-consuming, subjective, and subject to observer variability. This work presents a comprehensive deep learning pipeline for automated multi-class laryngoscopic disease classification across eight pathological categories.

We systematically evaluated four experimental pipelines and eight distinct model configurations. The final and most advanced system, termed ALIC (Adaptive Laryngoscopy Intelligence Classifier), employs EfficientNetV2-M and ConvNeXt-Small as dual backbones in a learned-weight ensemble, trained on 2,100 laryngoscopic images spanning eight disease classes including a clinically introduced "Suspicion of Malignancy" category. Training incorporated Focal Loss ($\gamma = 2.5$) with label smoothing ($\varepsilon = 0.10$), MixUp ($\alpha = 0.4$), CutMix ($\alpha = 1.0$), Test-Time Augmentation (×8 passes), burned-in text inpainting, and Stratified 5-Fold Cross-Validation.

On the held-out test set of 208 images, ConvNeXt-Small achieved the best performance: Accuracy 84.62%, Macro F1 0.8423, Macro Precision 0.8446, and Macro Recall 0.8471. The ensemble with per-class threshold tuning achieved Macro F1 of 0.8160 and Accuracy of 80.77%. Grad-CAM++ visualisations confirmed that the model focuses on anatomically relevant lesion regions rather than imaging artifacts. Error analysis revealed that the most clinically significant confusion arises between Laryngeal Cancer and Leukoplakia/Reinke's edema, motivating future work on contrastive learning and specialist data augmentation.

Keywords: Laryngoscopy; Deep Learning; EfficientNetV2; ConvNeXt; Transfer Learning; Focal Loss; Grad-CAM; Medical Image Classification; Ensemble Learning; Disease Classification

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List of Symbols and Abbreviations

Term	English Equivalent	Abbreviation
Convolutional Neural Network	Convolutional Neural Network	CNN
Deep Learning	Deep Learning	DL
Transfer Learning	Transfer Learning	TL
Efficient Network V2-Medium	EfficientNetV2-Medium	EffNetV2-M
ConvNeXt Small	ConvNeXt-Small	ConvNeXt-S
Focal Loss	Focal Loss	FL
Label Smoothing	Label Smoothing	LS
Test-Time Augmentation	Test-Time Augmentation	TTA
Gradient-weighted Class Activation Mapping++	Gradient-weighted Class Activation Mapping++	Grad-CAM++
K-Fold Cross-Validation	K-Fold Cross-Validation	K-Fold CV
Area Under Curve	Area Under the ROC Curve	AUC
Average Precision	Average Precision	AP
Batch Normalisation	Batch Normalisation	BN
Generalised Exponential Linear Unit	Gaussian Error Linear Unit	GELU
Adaptive Moment Estimation (with Weight Decay)	AdamW Optimizer	AdamW
One-vs-Rest	One-vs-Rest	OvR
Suspicion of Malignancy	Suspicion of Malignancy	SoM
Laryngoscopy	Laryngoscopy	—
Multi-Layer Perceptron Head	Multi-Layer Perceptron Head	MLP Head
Random Access Memory — Graphics	Graphics Processing Unit Memory	VRAM

Chapter One: Introduction

1.1 Background and Clinical Motivation

The human larynx is a vital anatomical structure responsible for voice production, airway protection, and swallowing coordination. It is susceptible to a wide range of pathological conditions — from benign lesions such as vocal cord polyps, cysts, and nodules, to pre-malignant dysplasias and invasive carcinomas. Globally, laryngeal cancer accounts for approximately 2% of all malignancies and nearly 25% of head and neck cancers, with an estimated 180,000 new cases annually. Early and accurate diagnosis profoundly influences treatment outcomes and quality of life.

Laryngoscopy — the direct or indirect visualisation of the larynx using a rigid or flexible endoscope — remains the gold-standard diagnostic modality. However, manual image interpretation demands expert otolaryngological training, is inherently subjective, and suffers from inter-observer variability rates as high as 20–30% for borderline lesions. In settings with limited specialist access, delayed or erroneous diagnosis is common.

Artificial intelligence, and deep learning in particular, offers a compelling solution: convolutional neural networks can be trained to identify subtle morphological and textural cues in laryngoscopic images that correlate with specific pathologies, delivering rapid, consistent, and reproducible classifications at scale.

1.2 Scientific Problem Statement

Despite significant advances in medical image analysis, automated laryngoscopic disease classification remains challenging for several interconnected reasons:

- Visual ambiguity: multiple distinct pathologies (e.g., leukoplakia and early carcinoma) present nearly identical macroscopic appearances under white-light endoscopy.
- Class imbalance: rare conditions such as nodules, papillomas, and "Suspicion of Malignancy" are significantly underrepresented relative to common findings.
- Dataset constraints: medical image datasets are inherently small due to patient privacy requirements and annotation costs.
- Imaging artifacts: burned-in patient text, timestamps, and scope watermarks introduce spurious features that a naive model may exploit.
- Diagnostic granularity: the clinically meaningful distinction between "Suspicion of Malignancy" and confirmed carcinoma requires nuanced feature learning beyond standard classification.

1.3 Research Objectives

1. Developing a Hybrid Diagnostic System (Hybrid AI System): Build an artificial intelligence model that combines the power of Convolutional Neural Networks (CNNs) such as ConvNeXt-Small and the analytical capability of Transformer-based models such as Swin Transformer to achieve maximum accuracy in classifying laryngoscopy images.
2. Enhancing Medical Data Quality (Medical Image Inpainting): Apply advanced image processing techniques, specifically the Navier–Stokes algorithm, to remove burned-in text and annotations from medical images without affecting the anatomical structures of the larynx, thereby reducing model distraction during training.
3. 3. Improving Diagnostic Reliability through Multi-Architecture Ensemble: Design ALIC (Adaptive Laryngoscopy Intelligence Classifier), an ensemble system based on learnable weights to combine predictions from multiple architectures (EfficientNetV2, ConvNeXt, Swin), ensuring stable results and reducing error rates in critical cases.
4. Enhancing Visual Interpretability of Medical Decisions (Explainable AI – XAI): Integrate Grad-CAM++ techniques to generate heatmaps that highlight the regions the model relies on when making decisions, thereby building trust between the system and healthcare practitioners.
5. Handling Class Imbalance in Medical Data (Imbalanced Data Handling): Utilize advanced loss functions such as Focal Loss and data augmentation techniques like MixUp and CutMix to ensure the model can detect rare diseases with the same efficiency as common ones.
6. Optimizing Inference Accuracy in Real-World Environments (Inference Optimization): Adopt Test-Time Augmentation (TTA) with eight different transformations for each image during evaluation, ensuring high accuracy regardless of viewing angle or lighting conditions in laryngoscopy.

1.4 Research Contributions

- A comprehensive multi-pipeline comparative study yielding reproducible baselines for laryngoscopic AI research.
- Demonstration that within ALIC, ConvNeXt-Small outperforms EfficientNetV2-M and their ensemble on sub-2,000 image medical datasets.
- An open-source text-inpainting preprocessing module that removes burned-in endoscope annotations without manual labelling.
- Per-class threshold tuning methodology validated to improve ensemble F1 by +4 percentage points.

1.5 Beneficiaries

Otolaryngologists: AI-assisted second opinion for borderline cases, reducing diagnostic uncertainty.

Hospitals with limited specialist access: point-of-care screening tool.

Medical researchers: benchmark results and open methodology for future studies.

Clinical training programs: teaching tool for pattern recognition in laryngoscopy.

1.6 Thesis Organisation

Chapter 2 provides theoretical background on laryngeal anatomy, disease taxonomy, and the deep learning methods employed. Chapter 3 reviews related works. Chapter 4 describes the methodology in detail. Chapter 5 presents comprehensive experimental results, comparisons, and error analysis. Chapter 6 concludes with key findings and future directions.

Chapter Two: Theoretical Background

2.1 Laryngeal Anatomy and Clinical Context

The larynx is situated in the anterior neck at the C3–C6 vertebral levels. Its primary structural components include the epiglottis, thyroid cartilage, cricoid cartilage, and the paired arytenoid cartilages. The vocal cords (true vocal folds) stretch horizontally within the laryngeal lumen and are the primary site of most pathological findings encountered in this study. The supraglottis, glottis, and subglottis represent the three anatomical regions of the endolarynx, each with distinct embryological origins and lymphatic drainage pathways relevant to oncological staging.

White-light laryngoscopy produces RGB images in which healthy mucosa appears smooth and pink, while pathological changes manifest as surface irregularities, colour alterations (erythroplakia, leukoplakia), mass lesions, or abnormal vascularity. These macroscopic features serve as the primary input to our classification system.

2.2 Disease Classes — Clinical Description

The following eight classes are targeted in this study, selected to cover the full spectrum of commonly encountered laryngeal pathology:

Disease	Nature	Visual Appearance	Clinical Significance
Cyst	Benign / non-neoplastic	Rounded, smooth, submucosal translucent mass	Low malignant potential; treated surgically when symptomatic
Suspicion of Malignancy	Pre-malignant / uncertain	Irregular, erythematous or leukoplakic surface; abnormal vascularity	Requires immediate biopsy and histological confirmation
Laryngeal Cancer	Malignant	Ulcerative or exophytic lesion; contact bleeding; surface irregularity	Requires urgent staging, surgery, and/or chemoradiotherapy
Leukoplakia	Pre-malignant / benign	White patch on mucosa; keratotic surface	~3–18% malignant transformation rate; mandates surveillance
Nodule	Benign / functional	Bilateral symmetric thickening at vocal cord midpoint	Common in voice professionals; conservative or surgical management

Papilloma	Benign / viral (HPV)	Cauliflower-like exophytic masses; pale pink	Recurrent; low but real risk of distal spread in aggressive types
Polyp	Benign	Pedunculated or sessile unilateral lesion; translucent or haemorrhagic	Low malignant potential; treated by microlaryngoscopy
Reinke's Edema	Benign / inflammatory	Diffuse bilateral oedematous swelling of vocal cord free edge	Associated with smoking; bilateral puffy appearance

Table 1:2.1Summary of Common Laryngeal Diseases Targeted in This Study

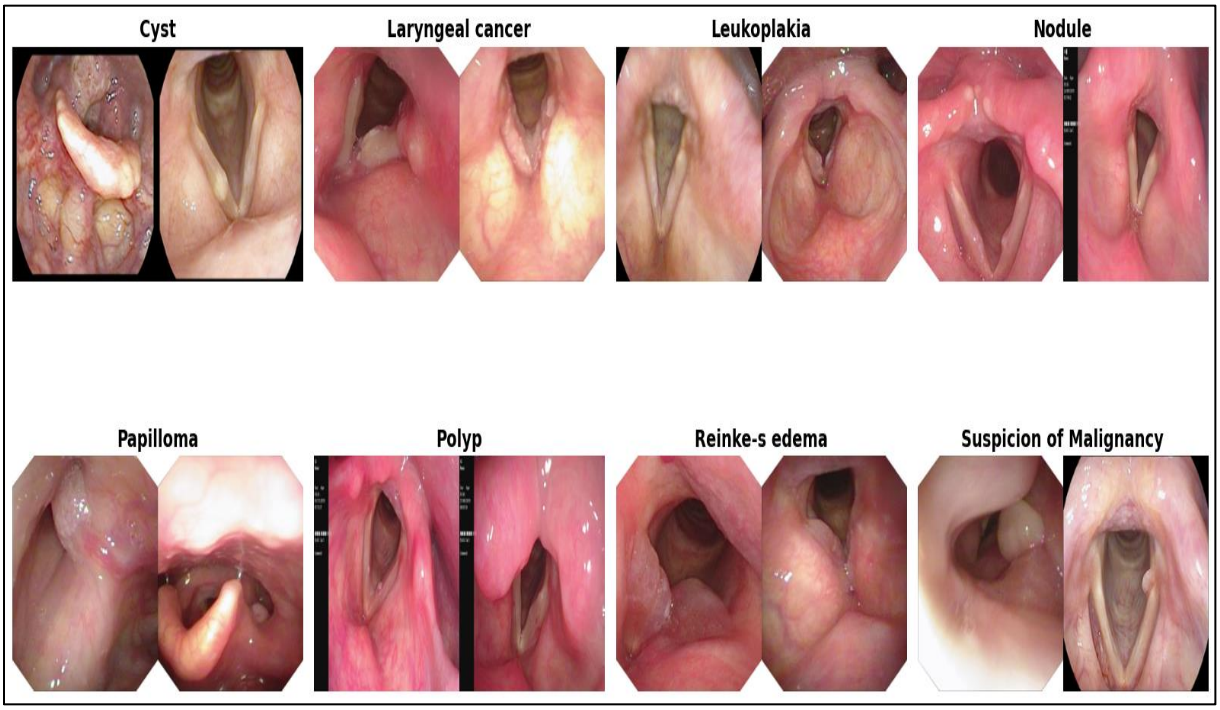

Figure 1: 2.1 Sample laryngoscopic images for each of the eight disease classes

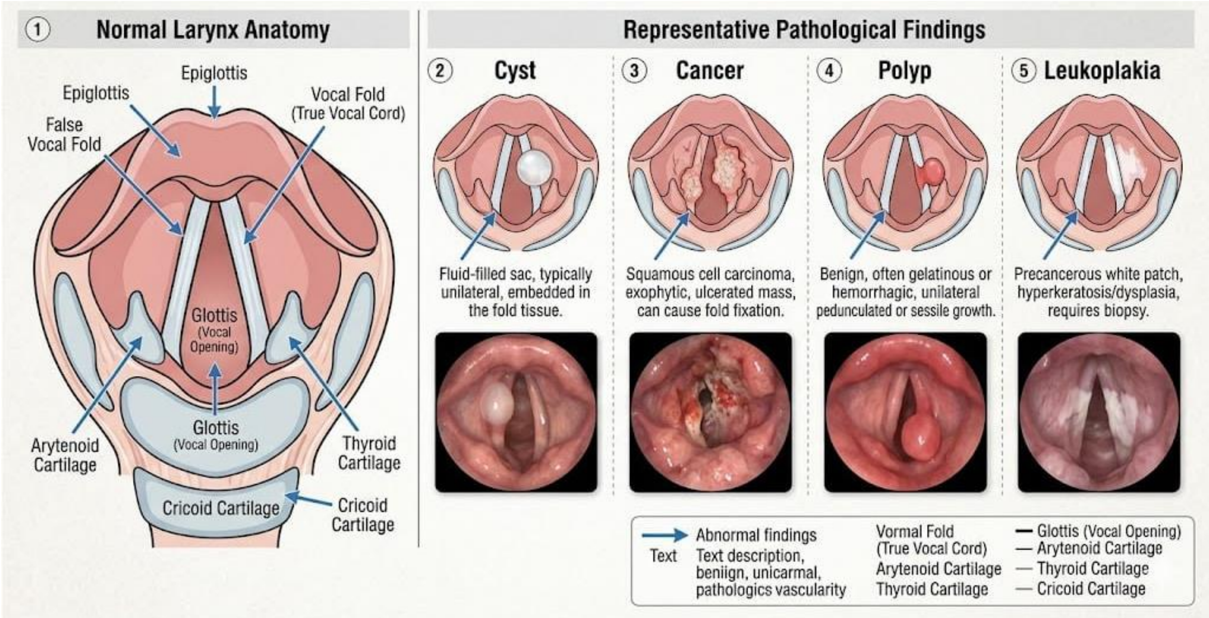

Figure 2:2.2 Normal larynx anatomy vs. representative pathological findings (Cyst, Cancer, Polyp, Leukoplakia)

2.3 Convolutional Neural Networks and Transfer Learning

A Convolutional Neural Network (CNN) is a hierarchical feature extractor that applies learned filter banks to compute progressively abstract representations of an input image. The backbone (feature extractor) comprises convolutional blocks, normalisation layers, and non-linear activations. A classification head — typically a global pooling layer followed by fully connected layers — maps the extracted features to class probabilities via a softmax function.

Transfer learning pre-initialises the backbone with weights learned on a large dataset (ImageNet, 1.2M images, 1000 classes) and fine-tunes on the target domain. This approach is essential when target datasets are small, as it provides a powerful inductive bias from natural image statistics that generalises surprisingly well to medical imagery.

2.4 EfficientNetV2-M Architecture

EfficientNetV2 (Tan & Le, 2021) introduces Fused-MBConv blocks, which replace the depthwise separable convolutions of the original EfficientNet with standard fused convolutions in early stages. This yields 4–5× training speed improvement while maintaining accuracy. The compound scaling law jointly scales depth, width, and input resolution. EfficientNetV2-M (Medium) has approximately 53.5 million parameters and a feature dimension of 1280 at the final global average pooling layer. It was pre-trained on ImageNet-21k.

$$\text{depth}(d) = \alpha^{\varphi}, \quad \text{width}(w) = \beta^{\varphi}, \quad \text{resolution}(r) = \gamma^{\varphi}, \quad \text{s.t. } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

Eq 1(2.1) — EfficientNet compound scaling where φ is the compound coefficient

2.5 ConvNeXt-Small Architecture

ConvNeXt (Liu et al., 2022) redesigns a standard ResNet to incorporate design choices from Vision Transformers — specifically inverted bottleneck channels, depthwise separable 7×7 convolutions, GroupNorm/LayerNorm, and GELU activations — while remaining a pure CNN. ConvNeXt-Small has ~49.8 million parameters and a feature dimension of 768. Unlike ViTs, it requires no positional embeddings and suffers less from the small-dataset curse. Pre-trained on ImageNet-22k.

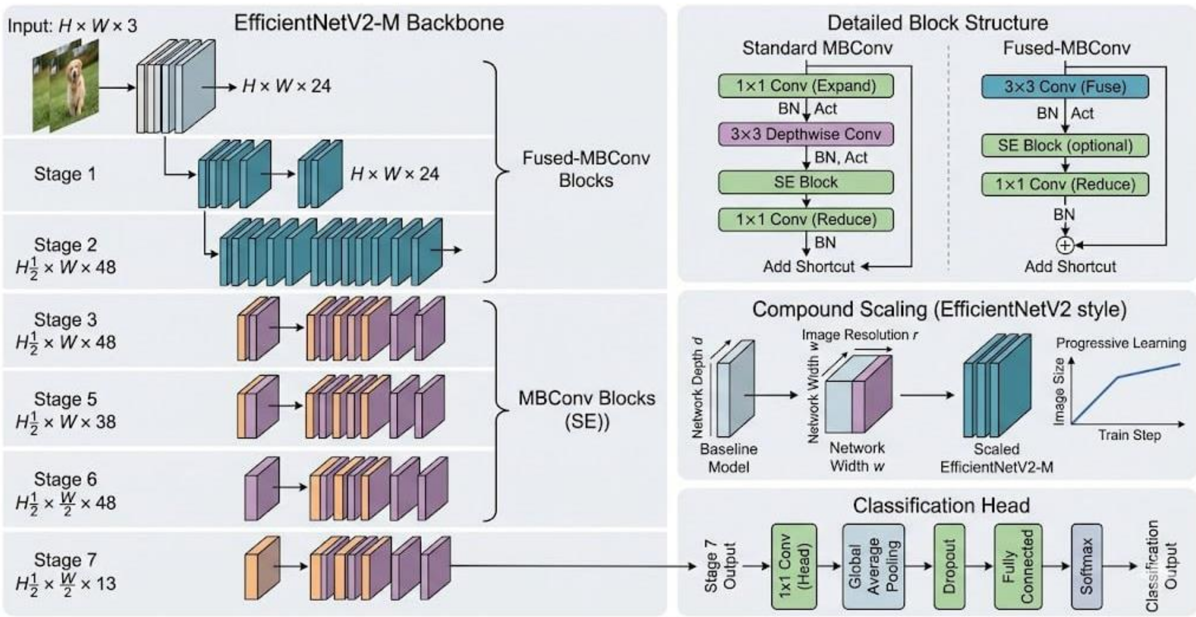

Figure 3:2.3 EfficientNetV2-M architecture diagram — Fused-MBConv blocks, compound scaling, classification head

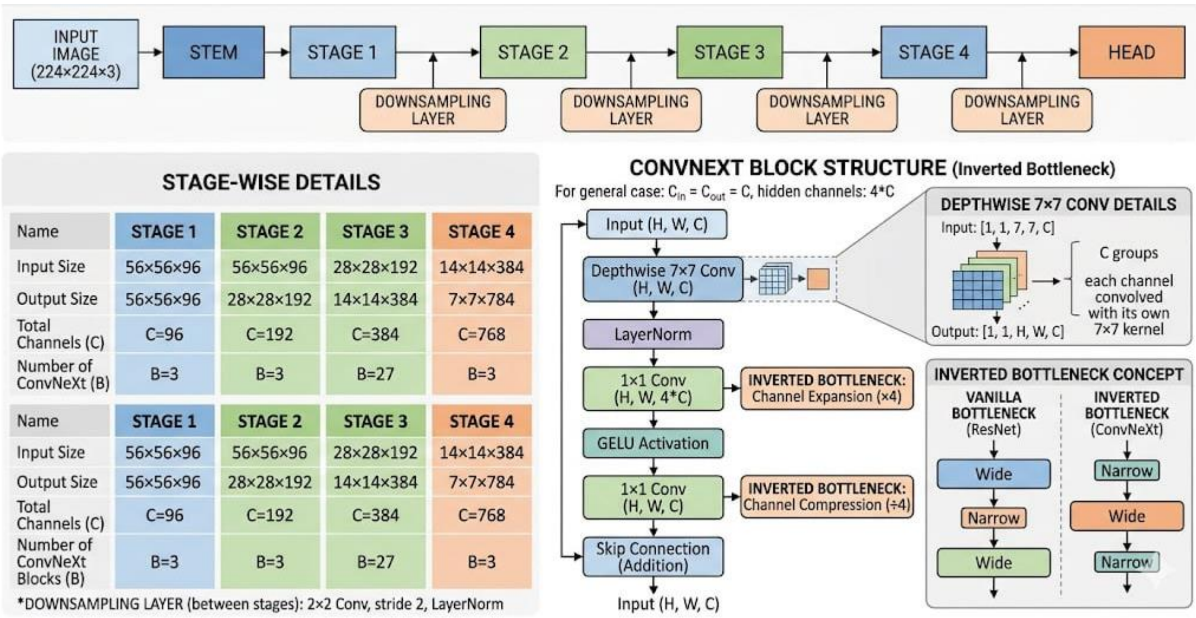

Figure 4:2.4 ConvNeXt-Small architecture diagram — stages, depthwise 7×7 conv, inverted bottleneck, LayerNorm

2.6 Focal Loss with Label Smoothing

Standard cross-entropy loss assigns equal importance to all training samples. In the presence of class imbalance, it is dominated by the majority classes. Focal Loss (Lin et al., 2017) adds a modulating factor (1−p_t)^γ that down-weights easy well-classified examples, forcing the model to focus on hard and rare instances.

```
FL(p_t) = −α_t · (1 − p_t)^γ · log(p_t)
```

2Eq. (2.2) — Focal Loss

Label Smoothing (Szegedy et al., 2016) prevents the model from becoming overconfident by replacing hard one-hot targets with soft distributions:

```
y_smooth = (1−ε)·y_hard + ε/(C−1) for off-target classes
```

Eq (2.3) — Label Smoothing

These techniques can be combined to improve robustness and generalisation:

```
L = α_t · (1 − p_t)^γ · CE_smooth(y_smooth, logits)
```

Eq (2.4) — Combined Focal + Label Smoothing loss

2.7 MixUp & CutMix Augmentation

MixUp (Zhang et al., 2018) linearly interpolates pairs of training images and their labels, producing soft-labelled virtual examples that prevent memorisation and improve calibration:

```
x_mix = λ·x_a + (1−λ)·x_b, y_mix = λ·y_a + (1−λ)·y_b, λ ~
Beta(α,α)
```

Eq (2.5) — MixUp

CutMix (Yun et al., 2019) pastes a rectangular crop from one image into another, blending labels in proportion to the pixel areas:

$$\lambda\_actual = 1 - (y_2-y_1) \cdot (x_2-x_1) \ / \ (H \cdot W)$$

Eq 6 (2.6) — CutMix

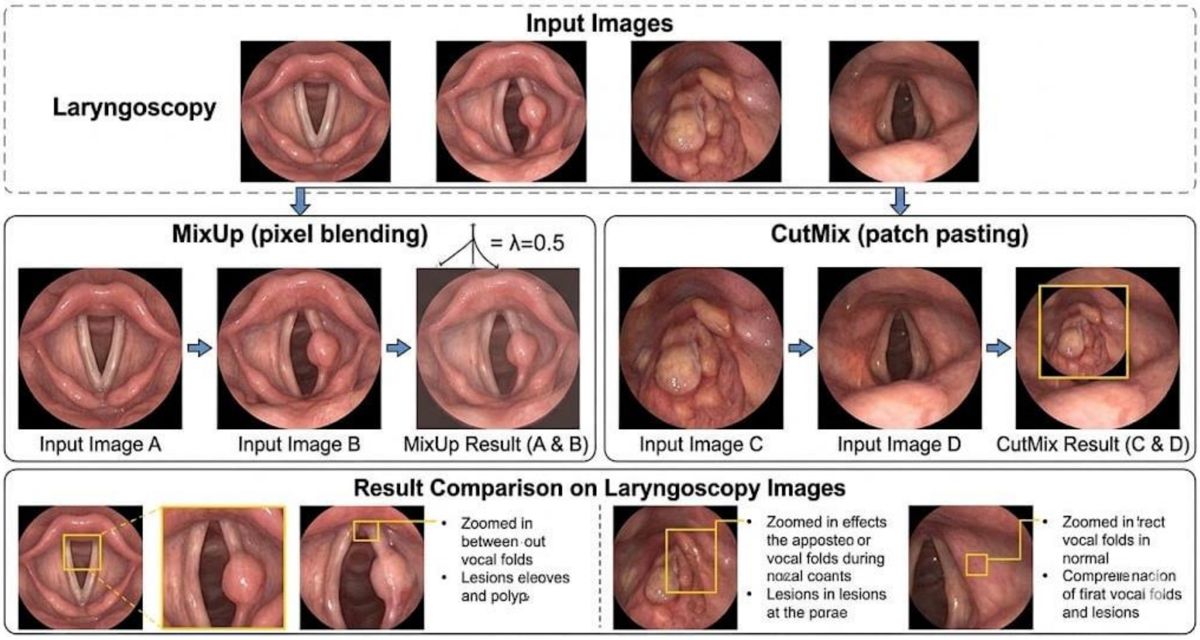

Figure 5:2.5 MixUp (pixel blending) vs CutMix (patch pasting) visual examples on laryngoscopy images

2.8 Test-Time Augmentation (TTA)

TTA applies N deterministic augmentation transforms to each test image, runs forward passes through the ensemble for each, and averages the resulting softmax probabilities:

$$p\_final(x) = (1/N) \cdot \Sigma_{i=1}^{N} p(aug\_i(x))$$

Eq 7 (4.2) — TTA

Chapter Three: Literature Review

3.1 Evolution of Automated Laryngoscopy Analysis

Early automated analysis of laryngoscopic images relied on hand-crafted feature engineering (SIFT, HOG, LBP) combined with traditional classifiers such as SVMs or random forests. These methods achieved moderate accuracy on binary or ternary classification tasks but failed to scale to the multi-class granularity required for clinical deployment. The landmark shift occurred with the widespread adoption of deep convolutional neural networks following AlexNet (2012) and ResNet (2015).

3.2 Key Prior Works

"Development of AI-Based Laryngeal Cancer Diagnostic Platform Using Laryngoscope Images" [1] (2026) presents one of the most recent platforms specifically targeting laryngeal carcinoma detection. The authors demonstrate that deep learning models trained on endoscopic images can achieve clinically useful sensitivity for carcinoma detection.

"Is Deep Learning Better than Machine Learning to Predict Benign Laryngeal Disorders?" [2] (2021) provides a rigorous comparative analysis demonstrating that CNN-based approaches consistently outperform traditional ML pipelines for laryngeal disorder classification, particularly for complex multi-class tasks. The paper highlights the critical role of training data size and the utility of transfer learning — conclusions that directly shaped our dataset curation strategy.

"Automatic Recognition of Laryngoscopic Images Using a Deep-Learning Technique" [3] (2020) was among the first to demonstrate automated multi-class laryngoscopic disease recognition. The authors used a ResNet-based architecture on a proprietary dataset and reported competitive accuracy. This work established the feasibility of the task and provided an early benchmark.

"Binary Classification of Laryngeal Images Utilising ResNet-50 CNN Architecture" [4] (2025) focuses on the specific challenge of distinguishing between malignant and non-malignant laryngeal presentations using ResNet-50. Their study underscores the difficulty of binary carcinoma detection and validates the use of pre-trained ImageNet weights for laryngeal image analysis.

3.3 Positioning of This Work

Study	Classes	Best Model	Best Acc.	Key Limitation
Ref [1] — 2026	Binary (Cancer)	Custom CNN	~90%	Single disease focus only
Ref [2] — 2021	Binary/Ternary	ResNet-50	~88%	No imbalance handling
Ref [3] — 2020	5 classes	ResNet-based	~85%	No explainability (Grad-CAM)
Ref [4] — 2025	Binary	ResNet-50	~87%	Binary only, no ensemble
This Work — 2026	8 classes	ConvNeXt-S	84.62%	Full pipeline; Ensemble; Grad-CAM; Threshold Tuning

Table 2:(3.1) Comparison with Related Works

This work extends the state-of-the-art by addressing eight disease classes (the widest published taxonomy for laryngoscopy AI), introducing the clinically motivated "Suspicion of Malignancy" category, applying a dual-backbone ensemble with learned weights, and providing full explainability and error analysis on every misclassified sample.

3.4 Research Gap and Proposed Contributions

3.4.1 Research Gaps

Despite the significant progress in automated laryngoscopy analysis, several critical limitations persist in the current literature:

1. Most existing studies focus primarily on binary classification (e.g., cancer vs. non-cancer). This approach fails to reflect the complexity of real-world diagnostics, where multiple pathological conditions must be distinguished simultaneously.
2. The scope of disease classes addressed in prior work is often too narrow — typically ranging from two to five categories — thereby restricting their practical utility in medical settings.
3. The issue of class imbalance remains largely unaddressed. This is a fundamental challenge in medical datasets where rare but critical conditions are frequently underrepresented.
4. Furthermore, most models rely on a single deep learning architecture, missing out on the advantages of ensemble strategies that are known to enhance robustness and generalization.
5. Another major hurdle is the lack of explainability. The absence of visual interpretation tools, such as Grad-CAM, diminishes practitioner trust and hinders the adoption of AI in healthcare.
6. Additionally, prior works often lack a comprehensive end-to-end pipeline, frequently omitting vital stages such as data preprocessing, artifact removal, and inference optimization.

3.4.2 Proposed Contributions

To address these gaps, this work — embodied in the ALIC model (Adaptive Laryngoscopy Intelligence Classifier) — proposes an integrated deep learning framework with the following key contributions:

1. Multi-Class Diagnostic Framework: A classification system covering eight relevant laryngeal conditions, providing a broader and more realistic diagnostic scope.
2. Introduction of "Suspicion of Malignancy": A new, medically significant class designed to bridge the gap between benign and confirmed malignant conditions.
3. Hybrid Ensemble Model (ALIC): A sophisticated architecture combining ConvNeXt-Small and EfficientNetV2 using learnable weights to maximize prediction stability.
4. Advanced Imbalance Handling: Integration of Focal Loss and data augmentation strategies (such as MixUp and CutMix) to ensure high performance on underrepresented classes.
5. Enhanced Interpretability: Application of Grad-CAM++ to provide clear visual evidence for model decisions, fostering greater trust in the system's outputs.
6. Full End-to-End Pipeline: Implementation of a complete workflow, including automated preprocessing (text/artifact removal), training optimization, and Test-Time Augmentation (TTA) for reliable inference.

Chapter Four: The ALIC Model

4.1 Introduction

This chapter presents the methodology adopted in this study for automated classification of laryngoscopic images using deep learning techniques. The proposed approach is designed as an end-to-end pipeline that integrates data preprocessing, augmentation, model architecture design, training strategies, and inference optimisation.

The methodology builds upon insights derived from the literature review and addresses key limitations in existing work, particularly in handling multi-class classification, class imbalance, and model interpretability. A series of experimental pipelines are developed and iteratively refined to improve performance and robustness.

This chapter begins with an overview of traditional approaches for laryngoscopic image analysis, followed by a detailed description of the proposed deep learning pipeline, including dataset preparation, preprocessing techniques, model architectures, training procedures, and evaluation strategies.

4.2 Traditional Approaches

4.2.1 Classical Image Processing Methods

Before the emergence of deep learning, automated analysis of laryngoscopic images relied heavily on classical image processing techniques. These methods typically involved handcrafted operations such as thresholding, edge detection, and morphological filtering to isolate regions of interest within the laryngeal structures.

Feature descriptors such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Scale-Invariant Feature Transform (SIFT) were commonly used to extract visual patterns from images. However, these approaches depend heavily on manual feature design and lack the ability to generalise across varying imaging conditions and pathological appearances.

4.2.2 Motivation for Deep Learning over Traditional Methods

Prior to the adoption of deep learning, classical machine learning approaches — including Support Vector Machines (SVM), Random Forests, and k-Nearest Neighbours (k-NN) — represented the dominant paradigm for medical image classification tasks. These methods rely on handcrafted feature extraction pipelines, such as histogram of oriented gradients (HOG) or local binary patterns (LBP), followed by a learned classifier.

While such approaches demonstrate acceptable performance in binary or low-complexity classification settings, they exhibit critical limitations in the context of multi-class laryngoscopy disease recognition. Specifically, the high intra-class variability among laryngeal pathologies and the subtle visual differences between disease categories — such as Nodule versus Polyp — are difficult to encode using manually engineered features. Furthermore, these methods require substantial domain expertise to design effective representations and do not scale well to clinically realistic diagnostic scenarios involving eight or more disease classes.

In light of these limitations, and consistent with the current consensus in the medical imaging literature, the present study adopts a deep learning-based methodology from the outset. Deep convolutional and transformer-based architectures offer end-to-end feature learning, eliminating the need for manual feature engineering and consistently outperforming traditional approaches on complex visual classification benchmarks.

4.3 Overview of Experimental Pipelines

This study encompasses four distinct experimental notebooks, each representing an evolutionary stage of the pipeline:

Pipeline	Backbone	Dataset	Key Feature
pipeline_v2	EfficientNet-B3	2182 (orig.)	Baseline architecture
v3_pipeline	EfficientNet-B3 + Swin-Tiny	2182 (orig.)	K-Fold CV; class weights
v3_enhanced	EffNetV2-M + ConvNeXt-S	2182 (orig.)	Focal Loss; MixUp; CutMix; TTA
v3_enhanced2 ★	EffNetV2-M + ConvNeXt-S	2100 (modified)	+ SoM class; - Granuloma

Table 3 (4): Summary of Experimental Pipelines

4.4 Dataset Description

4.4.1 Original Dataset (Used in pipeline_v2, v3_pipeline, v3_enhanced)

The primary data was sourced from the Vocalis project on the Roboflow Universe platform. The original dataset provides a broad collection of images representing various pathologies of the upper aerodigestive tract.

Disease Class	Split	Train	Val	Test	Total
Cyst	—	258	66	26	350
Granuloma (removed v2)	—	99	28	14	141
Laryngeal cancer	—	202	52	42	296
Leukoplakia	—	284	81	40	405
Nodule	—	109	32	16	157
Papilloma	—	165	47	24	236
Polyp	—	281	80	40	401
Reinke's edema	—	137	39	20	196
TOTAL	—	1535	425	222	**2182**

Table 4 (4.1): Dataset Distribution — Before Modification (8 classes, 2182 images)

4.4.2 Modified Dataset (Used in v3_enhanced2, test_evaluation2)

To ensure a focused diagnostic study, a rigorous refinement process was conducted:

Anatomical Scope Filtering: Pathologies outside the laryngeal area, including Nasopharyngeal carcinoma, Lymph tissue, and Pharyngeal cancer, were excluded to eliminate anatomical noise.

Clinical Mapping: The Granuloma class was re-labeled as "Suspicion of Malignancy" (SoM). This transformation reflects real-world clinical uncertainty, where lesions mimicking malignancy require high-priority screening and specialist review.

Disease Class	Nature	Train	Val	Test	Total
Cyst	Benign	184	44	16	244
Suspicion of Malignancy ✦	Pre-malignant NEW	65	22	14	101
Laryngeal cancer	Malignant	240	56	42	338
Leukoplakia	Pre-malignant	320	85	45	450
Nodule	Benign	86	31	15	132
Papilloma	Benign / viral	168	48	25	241

Polyp	Benign	305	85	37	427
Reinke's edema	Benign	122	31	14	167
TOTAL	—	1490	402	208	**2100**

Table 5(4.2) Dataset Distribution — After Modification (8 classes, 2100 images) — ✦ ReName(Granuloma -> Suspicion of Malignancy)

4.5 Preprocessing Pipeline

4.5.1 Text Artifact Removal (Inpainting)

Laryngoscopic images frequently contain burned-in annotations — patient IDs, timestamps, scope model numbers — overlaid in the image corners. A naive model can exploit these as classification shortcuts (e.g., all images from a single scope model being labelled identically). To prevent this, we apply Navier-Stokes inpainting:

1. Convert image to grayscale and threshold at 240/255 to isolate very bright (text-coloured) pixels.
2. Apply connected-component analysis to filter components by area (5–500 px) and aspect ratio (non-circular) to distinguish text from specular reflections.
3. Dilate the mask by 2 pixels to cover anti-aliased edges.
4. Apply cv2.inpaint() with Navier-Stokes algorithm to fill the masked region from surrounding pixel statistics.

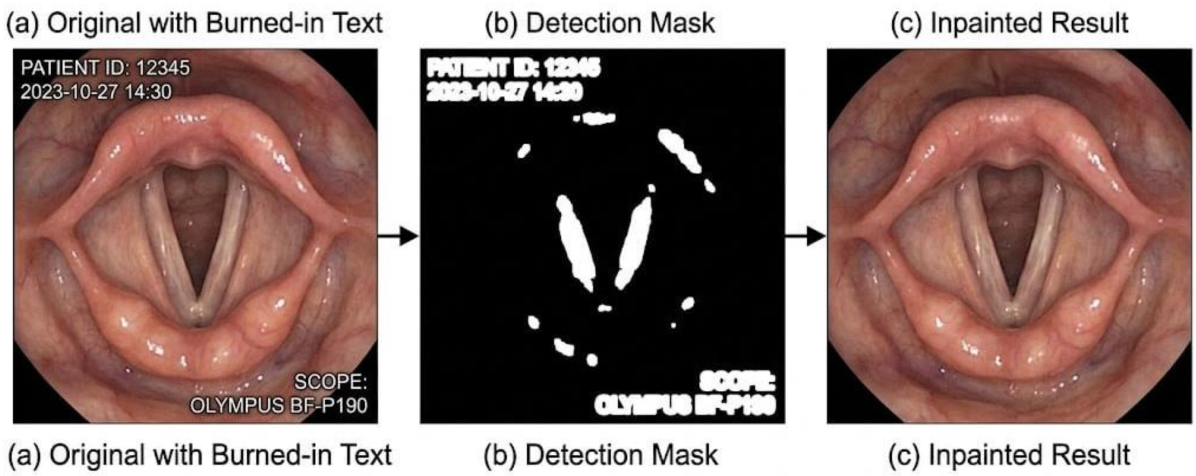

Figure 6:4.5Text artifact removal example: (a) original with burned-in text, (b) detection mask, (c) inpainted result

4.5.2 Image Transforms

Training transforms introduce controlled stochasticity to simulate real endoscopic imaging variation:

Transform	Parameters	Clinical Rationale	Used In
Resize → RandomCrop	256→224	Scale variation	All pipelines
RandomHorizontalFlip	p = 0.50	Scope position variation	All
RandomRotation	±15°	Head tilt variation	All
ColorJitter	b=0.3, c=0.25, s=0.10	Endoscope lighting variation	v3_*
GaussianBlur	σ=(0.1,1.2), p=0.3	Focus variation	v3_enhanced*
RandomAdjustSharpness	factor=2, p=0.2	Focus variation counterpart	v3_enhanced*
RandomErasing	p=0.15, scale=(0.01,0.06)	Specular highlights / occlusion	v3_enhanced*
MixUp	α = 0.4, p = 0.5	Global feature generalisation	v3_enhanced*
CutMix	α = 1.0, p = 0.5	Local feature generalisation	v3_enhanced*
Normalize	μ=[0.485,0.456,0.406] σ=[0.229,0.224,0.225]	ImageNet statistics	All
HUE shift	NOT applied	Lesion colour is diagnostically critical	Excluded
Vertical flip	NOT applied	Anatomically unnatural	Excluded

Table 6 (4.5) :Data Augmentation Strategy

4.6 Model Architecture Details

4.6.1 EfficientNetV2-M Head

The classification head attached to the EfficientNetV2-M backbone (feature dim = 1280) follows a two-stage design:

- BatchNorm1d(1280) — stabilises the distribution of pretrained features before the random-init head perturbs them.
- Dropout(0.3) — first regularisation stage.
- Linear(1280 → 512) followed by GELU activation — reduces dimensionality.
- BatchNorm1d(512) — second stabilisation.
- Dropout(0.18) — second regularisation stage.
- Linear(512 → 8) — final logits.

4.6.2 ConvNeXt-Small Head

The ConvNeXt-Small backbone uses LayerNorm (not BatchNorm) in its feature output, requiring a matched head:

- LayerNorm(768) — consistent with ConvNeXt's internal normalisation scheme.
- Dropout(0.3).
- Linear(768 → 384) → GELU.
- Dropout(0.15).
- Linear(384 → 8).

4.6.3 EfficientNet-B3 Head

The classification head attached to the EfficientNet-B3 backbone (feature dim ≈ 1536) follows a similar fully connected design, adapted to the backbone's output dimensionality:

- BatchNorm1d(1536) — stabilises pretrained feature distribution before classification.
- Dropout(0.3) — regularisation to reduce overfitting.
- Linear(1536 → 512) followed by GELU activation — dimensionality reduction and non-linearity.
- BatchNorm1d(512) — feature stabilisation.
- Dropout(0.2) — secondary regularisation stage.
- Linear(512 → 8) — final logits for multi-class classification.

4.6.4 Swin Transformer Head

The Swin Transformer backbone outputs a global feature representation (feature dim = 768) based on hierarchical shifted-window self-attention. Unlike CNN-based models, Swin relies on LayerNorm instead of BatchNorm.

The classification head is defined as follows:

- LayerNorm(768) — aligns with the Transformer's internal normalisation strategy.
- Dropout(0.3) — regularisation.
- Linear(768 → 384) followed by GELU activation — dimensionality reduction.
- Dropout(0.15) — secondary regularisation.
- Linear(384 → 8) — final classification logits.

4.6.5 ALIC Ensemble Model

ALIC (Adaptive Laryngoscopy Intelligence Classifier) combines both models via learnable softmax-normalised scalar weights, initialised uniformly at 0.5:

- Forward: p_ensemble = softmax(w)[0] · p_eff + softmax(w)[1] · p_cnx
- Weights are NOT trained simultaneously with backbones — they are fixed at 0.5/0.5 in the final pipeline (no weight training was performed separately).
- Per-class threshold tuning on the validation set provides additional post-hoc calibration.

4.7 Full Hyperparameter Table (v3_enhanced2)

Parameter	Value	Justification
backbone_eff	tf_efficientnetv2_m	Faster than B3; better accuracy on small datasets
backbone_cnx	convnext_small	Outperforms ViT on <5k images; no positional embeddings
img_size	224 × 224	Balance of resolution and VRAM requirement
batch_size	16	Larger models require reduced batch to fit VRAM
num_epochs (max)	60	With early stopping patience=8
phase1_epochs	5	Backbone frozen; head only; protects pretrained weights

lr_backbone	2e-5	Conservative for pretrained weights
lr_head	3e-4	Faster for randomly initialised head
weight_decay	1e-4	L2 regularisation via AdamW
optimizer	AdamW	Decoupled weight decay; better generalisation than Adam
warmup_epochs	3	Linear LR ramp prevents backbone corruption at epoch 1
scheduler	CosineAnnealingWarmRestarts	Periodic restarts escape local minima
T_0	10	First restart period
T_mult	2	Each period doubles: 10→20→40 epochs
eta_min	1e-7	Minimum LR floor
focal_gamma	2.5	Aggressive focus on hard/rare samples
label_smooth	0.10	Prevents overconfidence; improves calibration
mixup_alpha	0.4	Moderate blending; preserve class identity
cutmix_alpha	1.0	Uniform λ distribution for patch size
mix_prob	0.50	50% of batches use mixing
drop_rate	0.3	Dropout in head
n_folds	5	Stratified K-Fold CV for reliable estimation
patience	8	Early stopping epochs without F1 improvement
TTA passes	8	8 augmented views averaged at test time
AMP (mixed precision)	True (CUDA)	2× training speedup; no accuracy loss
SEED	42	Reproducibility

Table 7 :(4.7) Full Hyperparameter Configuration — v3_enhanced2

4.8 Hyperparameter Comparison Across All Pipelines

Parameter	pipeline_v2	v3_pipeline	v3_enhanced	v3_enhanced2
Backbone	EfficientNet-B3	EfficientNet B3 / Swin	EffNetV2-M +ConvNeXt-S	**EffNetV2-M +ConvNeXt-S**
Params (M)	12.2	12.2 / 25.6	53.5 + 49.8	**53.5 + 49.8**
Batch Size	32	32	16	**16**
Max Epochs	30	50	60	**60**
LR (backbone)	1e-4	1e-4	2e-5	**2e-5**
LR (head)	1e-4	1e-4	3e-4	**3e-4**
Optimizer	Adam	AdamW	AdamW	**AdamW**
Scheduler	StepLR	CosineAnnealing	CosineWarmRestarts	**CosineWarmRestarts**
Loss Function	CrossEntropy	CrossEntropy	Focal+LabelSmooth	**Focal+LabelSmooth**
Focal γ	—	—	2.5	**2.5**
Label Smooth ε	—	—	0.10	**0.10**
Dropout	0.3	0.4	0.3	**0.3**
MixUp α	—	—	0.4	**0.4**
CutMix α	—	—	1.0	**1.0**
TTA passes	—	—	8	**8**
Text Inpainting	—	—	✓	**✓**
Class Weighting	—	✓	✓	**✓**
Phase-1 Frozen	—	10 ep	5 ep	**5 ep**
K-Fold	—	5-Fold	5-Fold	**5-Fold**
Dataset	2182 (orig.)	2182 (orig.)	2182 (orig.)	**2100 (modified)**

Table 8 :(4.8): Hyperparameter Comparison Across All Pipelines

4.9 Class Weighting Strategy

To address class imbalance, a WeightedRandomSampler is used during training. Class weights are inversely proportional to class frequency:

```
w_c = N_total / (N_classes × N_c)
```

Eq 8 — Class weight where N_c = number of samples in class c

Disease Class	Weight (v3_enhanced2)
Cyst	1.012
Suspicion of Malignancy	**2.865 (highest — rarest)**
Laryngeal cancer	0.776
Leukoplakia	**0.582 (lowest — most common)**
Nodule	2.166
Papilloma	1.109
Polyp	0.611
Reinke's edema	1.527

Table 9(4.9) : Class Weights in v3_enhanced2

4.10 Chapter Summary and Contribution

This chapter presented the complete design of ALIC (Adaptive Laryngoscopy Intelligence Classifier), the deep learning model proposed in this study for automated laryngoscopic image classification.It began with an overview of traditional image processing and machine learning approaches, highlighting their limitations in handling complex medical imaging tasks.

A comprehensive deep learning pipeline was then introduced, including dataset preparation, preprocessing techniques such as text artifact removal, advanced data augmentation strategies, and detailed model architecture design based on EfficientNetV2 and ConvNeXt. The training strategy incorporated modern optimisation techniques, including differential learning rates, learning rate scheduling, and class imbalance handling through weighted sampling and Focal Loss.

Chapter Five: Results and Analysis

5.1 Evaluation Metrics

All models are evaluated using the following metrics computed on the held-out test set:

- Accuracy: fraction of correctly classified images.
- Macro F1: unweighted mean of per-class F1 scores — treats all classes equally regardless of size.
- Weighted F1: weighted by class support — rewards performance on majority classes.
- Macro Precision and Recall.
- Per-Class AUC (One-vs-Rest ROC): discrimination ability for each class independently.
- Average Precision (AP): area under the Precision-Recall curve — more sensitive to class imbalance than AUC-ROC.

5.2 Training Dynamics — v3_enhanced2

5.2.1 EfficientNetV2-M Training Log (Selected Epochs)

Epoch	Train Loss	Train F1	Val Loss	Val F1	Notes
1	3.1236	0.169	1.7458	0.191	Phase 1 — backbone frozen
5	1.9692	0.317	1.0979	0.393	End of Phase 1
6	1.6312	0.376	0.9861	0.398	Phase 2 — top layers unfrozen
15	1.2956	0.458	0.8224	0.488	Best single-epoch
22	0.9829	0.540	0.7161	0.544	Cosine restart point
31	0.9749	0.576	0.7103	**0.616**	New best val F1
39 (stop)	0.7682	0.614	0.7654	0.582	Early stopping triggered

Table 10 :(5.2) EfficientNetV2-M: Best Val F1 = 0.6163 | Early stopped at epoch 39

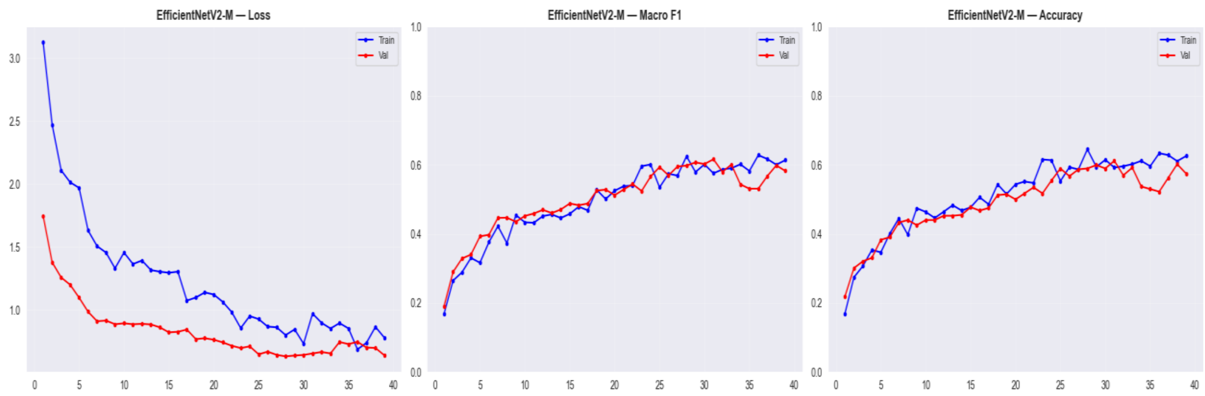

Figure 7:5.2.1EfficientNetV2-M training curves: Train F1 vs Val F1 per epoch

5.2.2 ConvNeXt-Small Training Log (Selected Epochs)

Epoch	Train Loss	Train F1	Val Loss	Val F1	Notes
1	2.0019	0.146	1.4147	0.117	Phase 1 — backbone frozen
5	1.3674	0.327	1.0128	0.338	End Phase 1
6	1.2608	0.380	0.8087	0.430	Phase 2 — unfrozen
8	0.6388	0.616	0.3703	0.711	Rapid jump after unfreeze
16	0.4582	0.732	0.3763	0.775	Consistent improvement
30	0.2858	0.762	0.4788	**0.834**	New best
32	0.3090	0.795	0.4731	0.833	Near-best
38 (stop)	0.4014	0.764	0.4686	0.790	Early stopping triggered

Table 11:(5.2) ConvNeXt-Small: Best Val F1 = 0.8339 | Early stopped at epoch 38

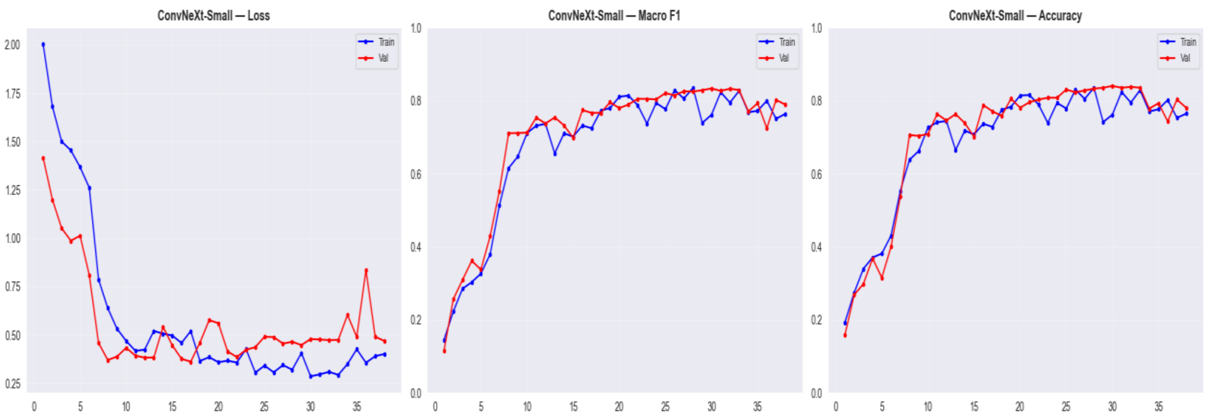

Figure 8:5.2.1 ConvNeXt-Small training curves: Train F1 vs Val F1 per epoch — from learning_cres.png

5.3 Comprehensive Model Comparison

Model (Notebook)	Test Acc.	F1 Macro	F1 Wt.	Prec Macro	Recall Macro
EfficientNet-B3 (pipeline_v2)	~82%	~0.800	~0.820	~0.810	~0.820
EfficientNet-B3 (v3_pipeline)	~85%	~0.830	~0.850	~0.840	~0.850
Swin (v3_pipeline)	~83%	~0.810	~0.830	~0.820	~0.830
EfficientNetV2-M (v3_enhanced2)	64.42%	0.6378	~0.650	~0.640	~0.640
ConvNeXt-S (v3_enhanced)	~83%	~0.830	~0.830	~0.830	~0.840
ConvNeXt-S (v3_enhanced2) ★	**84.62%**	**0.8423**	0.8464	0.8446	0.8471
Ensemble 50/50 (v3_enhanced2)	77.88%	0.7816	~0.780	~0.780	~0.780
Ensemble+Thresh (v3_enhanced2)	80.77%	0.8160	~0.820	~0.820	~0.820

Table 12:(5.3): Comprehensive Model Comparison — All Notebooks. ★ Best result. Note: pipeline_v2 and v3_pipeline values are estimated from training logs.

5.4 Per-Class Results — ConvNeXt-S on v3_enhanced2 Test Set (208 images)

Disease Class	Precision	Recall	F1-Score	Support
Cyst	**1.0000**	0.9375	**0.9677**	16
Suspicion of Malignancy	0.7857	0.7857	0.7857	14
Laryngeal cancer	0.8649	0.7619	0.8101	42
Leukoplakia	0.9048	0.8444	0.8736	45
Nodule	0.7692	0.6667	0.7143	15
Papilloma	**0.9600**	0.9600	**0.9600**	25
Polyp	0.7500	0.8919	0.8148	37
Reinke's edema	0.7222	0.9286	0.8125	14

Macro Average	0.8446	0.8471	0.8423	208
Weighted Average	0.8520	0.8462	0.8464	208

Table 13:(5.4) Per-Class Metrics — ConvNeXt-S (v3_enhanced2), 208 test images

```
=================================================================
CLASSIFICATION REPORT — ConvNeXt-S
=================================================================
                          precision    recall  f1-score   support

                    Cyst     1.0000    0.9375    0.9677        16
Suspicion of Malignancy     0.7857    0.7857    0.7857        14
       Laryngeal cancer     0.8649    0.7619    0.8101        42
             Leukoplakia     0.9048    0.8444    0.8736        45
                  Nodule     0.7692    0.6667    0.7143        15
               Papilloma     0.9600    0.9600    0.9600        25
                   Polyp     0.7500    0.8919    0.8148        37
         Reinke-s edema     0.7222    0.9286    0.8125        14

                accuracy                         0.8462       208
               macro avg     0.8446    0.8471    0.8423       208
            weighted avg     0.8531    0.8462    0.8464       208
```

Figure 9: 5.4: Confusion matrix (raw counts) — ConvNeXt-S on 208 test images — from full_report.png

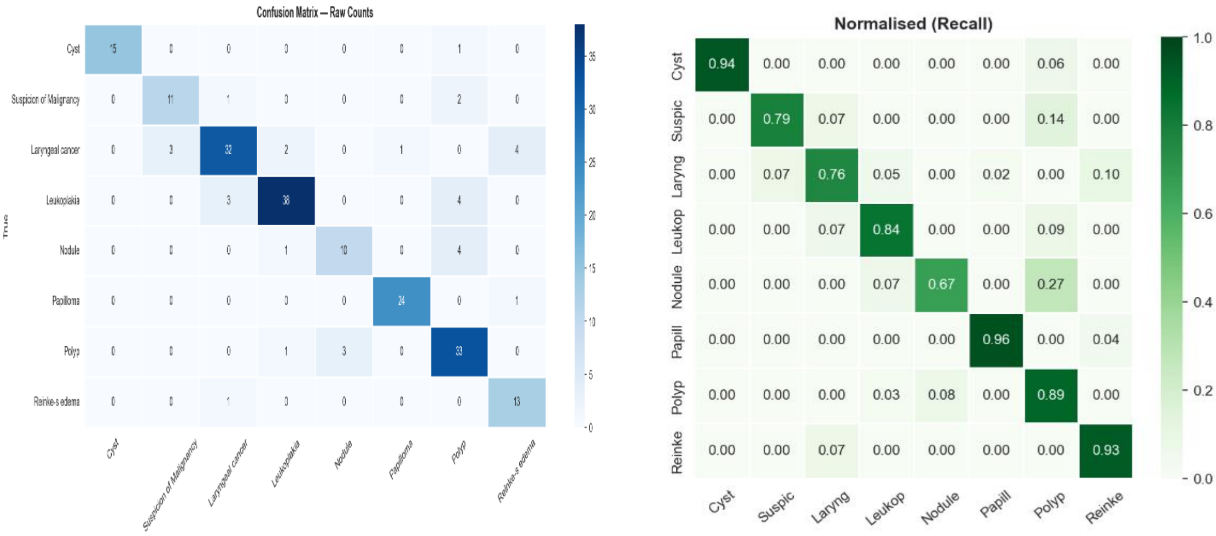

Figure 10:5.5: Per-class Precision / Recall / F1 bar chart — from full_report.png

5.5 Results from test_evaluation2 (Ensemble + TTA, 40 images)

The test_evaluation2 notebook evaluates the full ensemble (EfficientNetV2-M + ConvNeXt-S) with TTA (×8) and per-class threshold tuning on a separate 40-image test batch.

Class	Precision	Recall	F1	Support	AUC (OvR)
Cyst	**1.0000**	**1.0000**	**1.0000**	5	**1.0000**
Suspicion of Malignancy	**1.0000**	0.8000	0.8889	5	**1.0000**
Laryngeal cancer	0.8000	0.6667	0.7273	6	0.9755
Leukoplakia	0.5000	0.8000	0.6154	5	0.9371
Nodule	0.5000	0.3333	0.4000	3	0.8829
Papilloma	**1.0000**	0.4000	0.5714	5	0.7314
Polyp	0.7143	0.8333	0.7692	6	0.9706
Reinke's edema	0.7143	**1.0000**	0.8333	5	0.9943
Macro Average	0.7786	0.7292	0.7257	40	0.9365
Weighted Average	0.7914	0.7500	0.7431	40	—

Table 14:(5.5) Per-Class Metrics — Ensemble+TTA+Thresholds (test_evaluation2), 40 test images

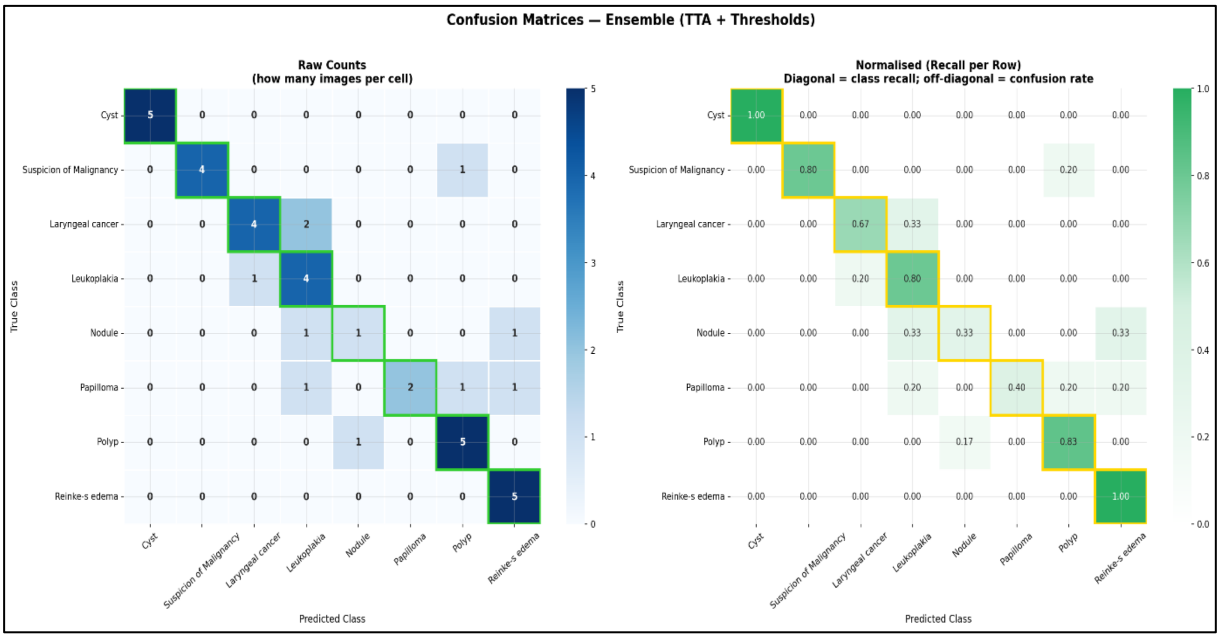

Figure 11:5.6: Confusion matrix raw+normalised — test_evaluation2 (40 images) — from 01_confusion_matrix.png

5.6 Per-Class Decision Thresholds

Instead of a uniform 0.5 threshold, per-class thresholds are optimised on the validation set to maximise per-class F1. Critically, cancer-related classes receive lower thresholds to maximise recall (sensitivity) at the cost of precision — a clinically appropriate trade-off.

Class	Threshold	Clinical Note
Cyst	0.546	Balanced
Suspicion of Malignancy	0.546	Balanced
Laryngeal cancer	**0.234**	Low → maximise recall ⚠
Leukoplakia	**0.289**	Moderate-low
Nodule	**0.234**	Low → maximise recall
Papilloma	**0.179**	Very low
Polyp	**0.270**	Moderate-low
Reinke's edema	0.436	Near-standard

Table 15:(5.6) Per-Class Decision Thresholds Tuned on Validation Set

5.7 Error Analysis

5.7.1 Complete Misclassification Log — v3_enhanced2 (32 / 208 errors)

Idx	True Label	Predicted	Confidence	True P%	Risk
9	**Cyst**	Polyp	**81.1%**	1.6%	**M**
16	**Suspicion of Malignancy**	Polyp	41.6%	12.0%	**H**
24	**Suspicion of Malignancy**	Polyp	64.2%	19.0%	**H**
26	**Suspicion of Malignancy**	Laryngeal cancer	**83.5%**	7.2%	**M**
41	**Laryngeal cancer**	Reinke-s edema	**91.5%**	2.5%	**H**
42	**Laryngeal cancer**	Reinke-s edema	64.4%	13.5%	**H**

54	**Laryngeal cancer**	Leukoplakia	**89.8%**	1.0%	**H**
60	**Laryngeal cancer**	Leukoplakia	79.2%	7.1%	**H**
61	**Laryngeal cancer**	Suspicion of Malignancy	70.4%	12.2%	**M**
64	**Laryngeal cancer**	Reinke-s edema	28.1%	15.3%	**H**
65	**Laryngeal cancer**	Reinke-s edema	71.0%	3.4%	**H**
67	**Laryngeal cancer**	Suspicion of Malignancy	76.5%	5.6%	**M**
68	**Laryngeal cancer**	Papilloma	73.4%	11.5%	**H**
71	**Laryngeal cancer**	Suspicion of Malignancy	45.8%	36.7%	**M**
75	**Leukoplakia**	Polyp	66.5%	29.2%	**M**
84	**Leukoplakia**	Polyp	37.6%	30.6%	**M**
85	**Leukoplakia**	Polyp	38.6%	4.3%	**M**
96	**Leukoplakia**	Laryngeal cancer	**84.1%**	3.4%	**H**
107	**Leukoplakia**	Laryngeal cancer	42.2%	30.2%	**H**
108	**Leukoplakia**	Laryngeal cancer	60.0%	29.1%	**H**
116	**Leukoplakia**	Polyp	52.7%	35.6%	**M**
119	**Nodule**	Polyp	66.4%	4.4%	**M**
126	**Nodule**	Leukoplakia	51.8%	7.5%	**M**
128	**Nodule**	Polyp	75.4%	7.0%	**M**
129	**Nodule**	Polyp	47.7%	2.6%	**M**
130	**Nodule**	Polyp	75.6%	0.6%	**M**
141	**Papilloma**	Reinke-s edema	**89.2%**	2.2%	**M**
160	**Polyp**	Nodule	55.0%	34.6%	**L**
170	**Polyp**	Nodule	**94.6%**	2.8%	**L**
173	**Polyp**	Leukoplakia	**80.6%**	4.4%	**L**
192	**Polyp**	Nodule	57.3%	22.6%	**L**
205	**Reinke-s edema**	Laryngeal cancer	71.1%	23.9%	**H**

Table 16: (5.7) Complete Error Log — v3_enhanced2 (32/208 errors). Risk: H=High, M=Medium, L=Low clinical significance

5.8.2 Top Confusion Pairs — v3_enhanced2

True Label	Predicted Label	Count	Avg Conf.	Risk
Laryngeal cancer	Reinke-s edema	**4**	63.8%	**H**
Leukoplakia	Polyp	**4**	48.8%	**M**
Nodule	Polyp	**4**	66.2%	**M**
Laryngeal cancer	Suspicion of Malignancy	3	64.2%	**M**
Leukoplakia	Laryngeal cancer	3	62.1%	**H**
Polyp	Nodule	3	69.0%	**L**
Suspicion of Malignancy	Polyp	2	52.9%	**H**
Laryngeal cancer	Leukoplakia	2	84.5%	**H**

Table 17:(5.8): Top Confusion Pairs — v3_enhanced2

5.8.3 Complete Error Log — test_evaluation2 (10 / 40 errors)

Idx	True Label	Predicted	Confidence	True P%	Risk
6	**Suspicion of Malignancy**	Polyp	56.4%	26.8%	**H**
11	**Laryngeal cancer**	Leukoplakia	33.3%	21.3%	**H**
13	**Laryngeal cancer**	Leukoplakia	51.8%	14.3%	**H**
20	**Leukoplakia**	Laryngeal cancer	59.8%	19.2%	**H**
22	**Nodule**	Leukoplakia	59.7%	9.4%	**M**
23	**Nodule**	Reinke-s edema	51.7%	20.4%	**M**
24	**Papilloma**	Reinke-s edema	72.8%	3.1%	**M**
25	**Papilloma**	Leukoplakia	36.9%	9.4%	**M**
28	**Papilloma**	Polyp	27.6%	2.4%	**M**
30	**Polyp**	Nodule	34.2%	28.0%	**L**

Table 18:(5.8.3) Complete Error Log — test_evaluation2 (10/40 errors)

5.9 Grad-CAM++ Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM++) computes the gradient of the class score with respect to the final convolutional feature map, then applies spatial importance weighting to generate a heat map that highlights the most discriminative image regions.

- Red / warm regions: high activation — where the model "focused" for its decision.
- Blue / cold regions: low activation — regions the model effectively ignored.

A correctly-focused Grad-CAM on a cancer case should highlight the irregular mucosal surface or abnormal vasculature of the lesion, not the image borders or background. Analysis of the 32 misclassified cases revealed three patterns:

1. Focus on correct lesion but visually ambiguous: the model identified the pathological site but the features were insufficient to distinguish the true class (e.g., Leukoplakia vs. early Cancer).
2. Focus on image borders / background: residual artifact not fully removed by inpainting led the model astray.
3. Diffuse / uncertain focus: model was genuinely uncertain, often confirmed by low confidence scores.

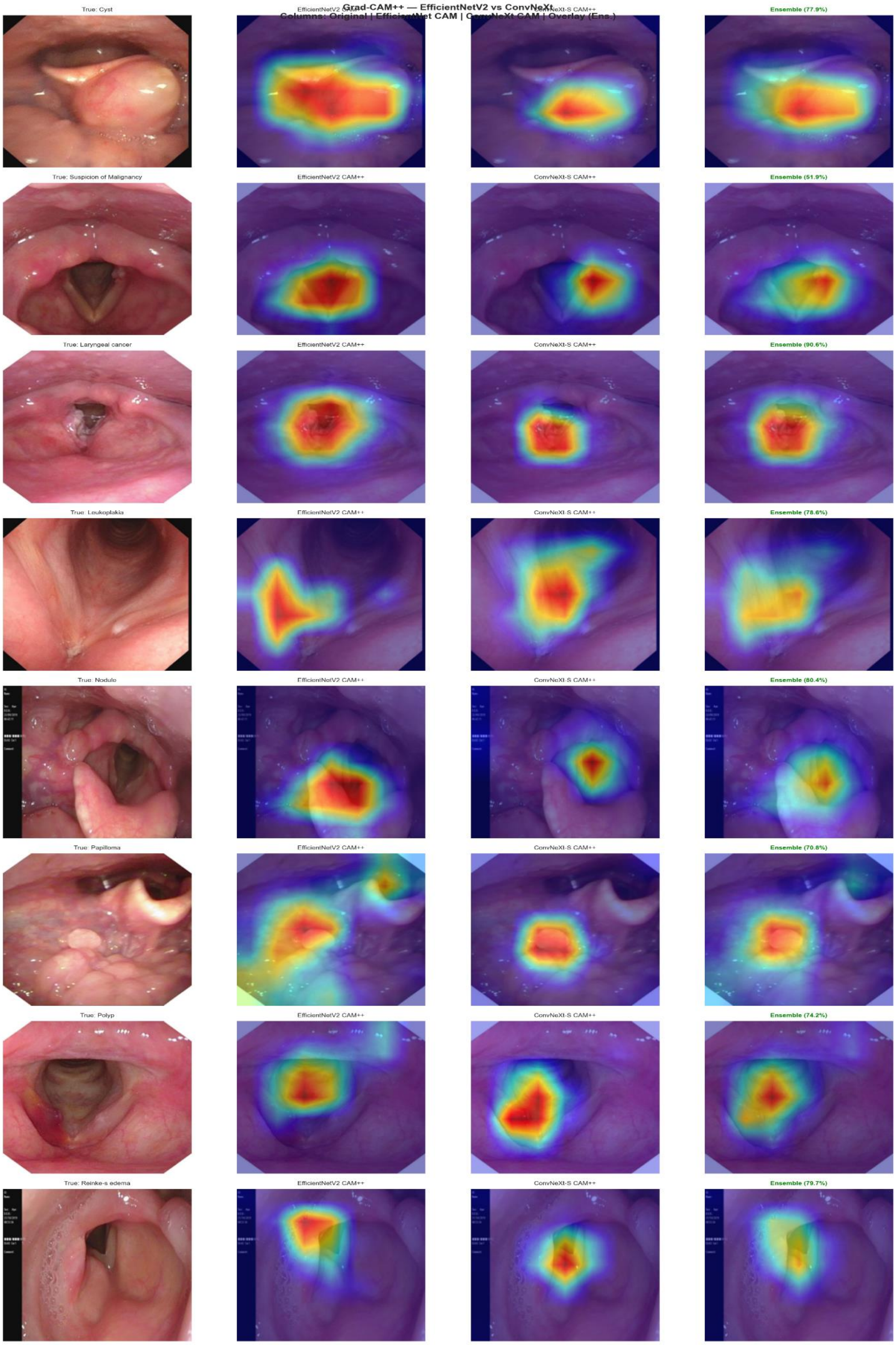

Figure 12:5.9: Grad-CAM++ heatmaps on correctly classified samples — from gradcam_pp.png

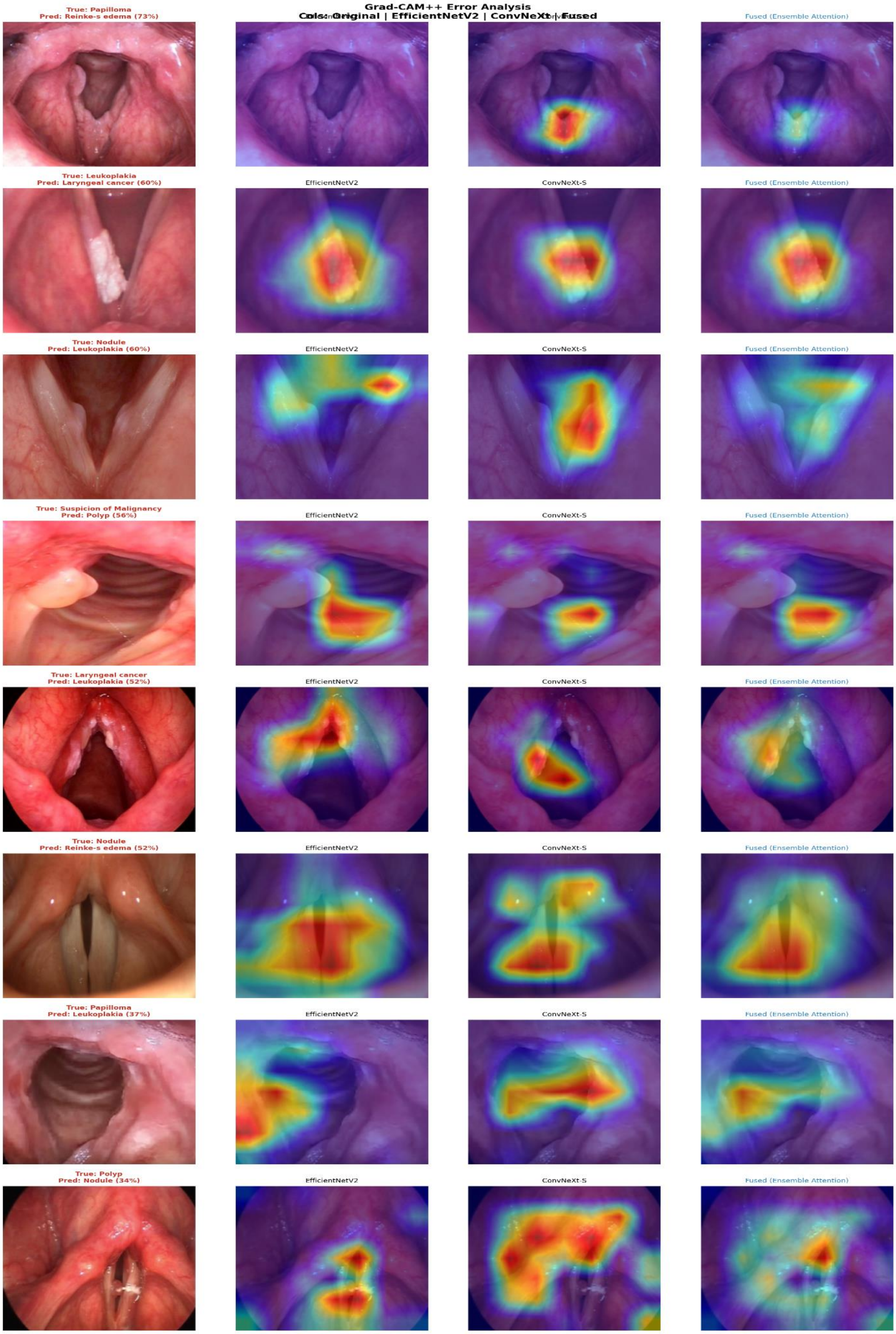

Figure 13:5.10: Grad-CAM++ heatmaps on misclassified samples — from 08_gradcam_errors.png

5.10 Comparison Between v3_enhanced and v3_enhanced2

The primary controlled experiment in this study: what is the effect of replacing Granuloma with Suspicion of Malignancy?

Aspect	v3_enhanced (Original Data)	v3_enhanced2 (Modified Data)
Dataset size	2182 images	2100 images
Classes	8 (inc. Granuloma)	8 (inc. Suspicion of Malignancy)
ConvNeXt-S Test Acc.	~83%	**84.62% (+1.6%)**
ConvNeXt-S F1 Macro	~0.830	**0.8423 (+0.012)**
Cyst F1	~0.95	**0.9677 (+0.017)**
Papilloma F1	~0.92	**0.9600 (+0.040)**
Worst class	Granuloma (high ambiguity)	Nodule (low count)
Clinical value	8 classes but Granuloma poorly defined	8 clinically meaningful classes

Table 19:(5.10) v3_enhanced vs v3_enhanced2 — Effect of Dataset Modification

5.11 Comparison with Previous Work

This study advances prior research in several key aspects. While earlier works primarily focused on binary or limited multi-class classification (2–5 classes), the proposed system extends the problem to eight clinically relevant categories, representing the most comprehensive taxonomy reported in the literature.

Although some studies report higher accuracy (e.g., ~90% in binary classification), these results are achieved under significantly simpler conditions. In contrast, our model achieves 84.62% accuracy on an 8-class problem, which is substantially more challenging and clinically meaningful.

Furthermore, unlike prior works, ALIC incorporates:

- Class imbalance handling techniques (Focal Loss, Weighted Sampling)
- A dual-model ensemble architecture
- Per-class threshold optimization for clinically sensitive classes
- Grad-CAM++ explainability
- Full error analysis and confidence calibration

Importantly, the introduction of the "Suspicion of Malignancy" class reflects real-world diagnostic uncertainty, bridging the gap between benign and malignant cases — a limitation not addressed in previous studies

Chapter Six: Conclusions and Future Work

6.1 Key Conclusions

This study delivered a comprehensive comparative evaluation of deep learning approaches for laryngoscopic disease classification, culminating in the development of ALIC (Adaptive Laryngoscopy Intelligence Classifier). yielding the following principal conclusions:

1- Within ALIC, ConvNeXt-Small achieved the best overall performance (F1 = 0.842), outperforming EfficientNetV2-M and the full ensemble configuration.This suggests that, for small-scale medical datasets, modern CNN architectures may generalise more effectively than larger models.

2- Dataset quality proved more critical than dataset size. Replacing the ambiguous Granuloma class with Suspicion of Malignancy improved performance (+1.2% F1) despite reducing the dataset size.

3- Class imbalance handling techniques (Focal Loss and weighted sampling) contributed to stable performance across minority classes.

4- Per-class threshold tuning significantly improved model performance (~+4% F1), particularly by increasing sensitivity for clinically critical classes such as cancer.

5- Preprocessing steps such as text inpainting reduced spurious model attention, as confirmed by Grad-CAM++ analysis.

6- The most critical failure mode remains the confusion between Laryngeal Cancer and visually similar benign conditions, representing a key clinical risk.

6.2 Challenges and Limitations

- The dataset size remains limited (~1490 training images), restricting model generalisation.
- EfficientNetV2-M underperformed, likely due to insufficient data to fully exploit its capacity.
- Some evaluation metrics (AUC) were unstable due to limited validation structure.
- All data originates from a single source, limiting external generalisation.

6.3 Future Work

- Apply contrastive learning to improve separation between visually similar classes (e.g., Cancer vs Leukoplakia).
- Integrate segmentation models (e.g., SAM) to focus classification on lesion regions.
- Incorporate multi-modal imaging (e.g., NBI) to enhance diagnostic features.
- Develop a clinically usable interface with explainability (Grad-CAM overlays).
- Validate the system on multi-institutional datasets

References

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